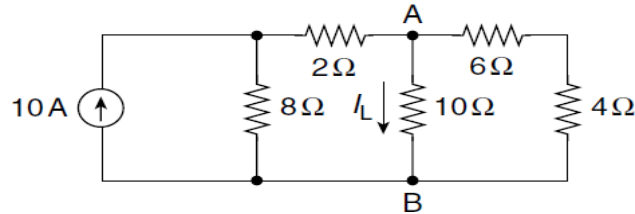
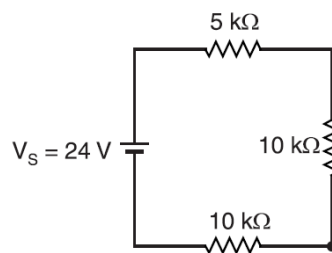


Tutorial - 3

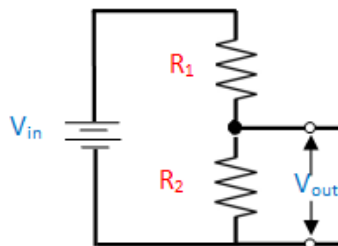
- Using Norton's theorem, calculate the current through the $10\ \Omega$ resistor in the network shown in Fig. 1.



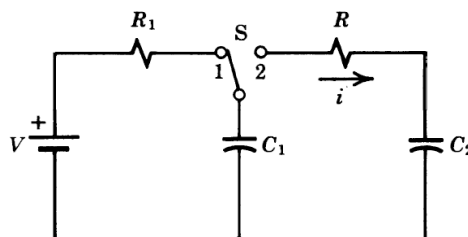
- Find voltage across the $10\ \text{k}\Omega$ resistor in the following circuit using voltage divider concept.



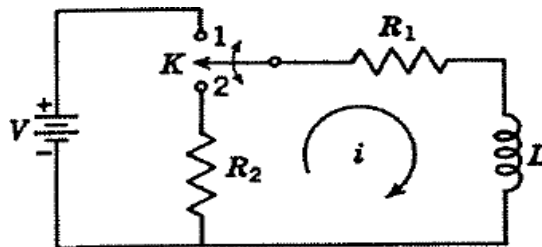
- In the following circuit, if desired output voltage V_{out} is 20 Volts, what value of R_2 should be used? Assume 50 Volts of input voltage is applied and $R_1 = 15\ \Omega$



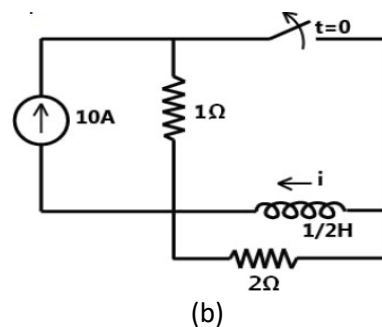
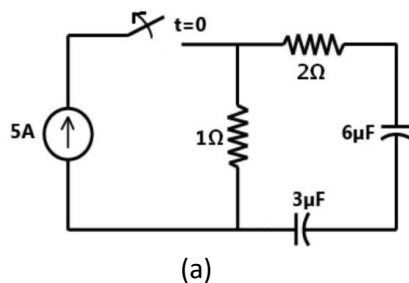
- In the circuit show below, the capacitor C_2 is uncharged, and switch S has been in position 1 for a long time.
 - Under steady-state condition, what is the current in C_1 ? The current in R_1 ? The voltage across C_1 ?
 - If switch S is thrown to position 2 at time $t = 0$, write the equations governing current i as a function of time.
 - Assume $C_1 = C_2$, solve for $i(t)$ and sketch $i(t)$ on a graph.



5. In the circuit shown below, the switch is changed from position 1 to position 2 at $t = 0$, a steady-state current having previously been established in the RL circuit, find the current in the circuit for $t \geq 0$.



6. Calculate time constants of the following circuits



7. In Fig. 3.0 switch S is opened at $t=0$ after being closed for a long time. Predict current i as a function of time after $t=0$. Sketch $i(t)$ just before $t = 0$ and for a significant time after.

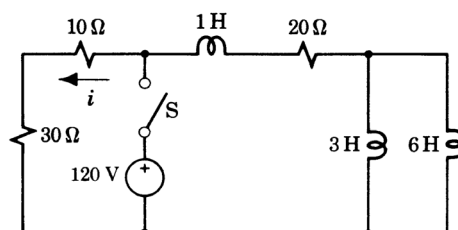
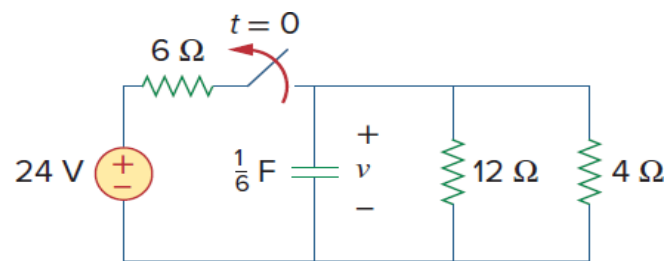


Fig. 3

8. Energy is to be stored in an inductor and dissipated in a 10Ω resistor. The power in the resistor must be cut in half every second.
- Design the inductor, i.e., specify L of an ideal inductance.
 - If a practical inductor introduces 0.5Ω of resistance for every $10H$ of inductance, specify L .
9. Two bicycle wheels appear to be similar in dimension and construction, but one has a time constant twice that of the other. Which one will be easier to pedal? Why?

10. For the circuit below, if switch opens at $t=0$, find $v(t)$ for $t \geq 0$



11. For the circuit below, find $i(t)$ for $t \geq 0$

